

Problem 7.11

You want to buy a sailboat when you retire 5 years from now. The sailboat you wish to buy costs \$50,000 today and inflation is at 4% per year. What do you need to put into an account today to have enough funds to be able to purchase the boat 5 years from now if the account earns 7% per year?

Solution

First, find the cost of the sailboat in 5 years using the inflation rate of 4%:

$$50,000(1.04)^5 = 60,832.64512.$$

So the boat is expected to cost

$$\$60,832.65$$

5 years from now.

Let X be the amount that must be invested today at 7% effective annually. Then

$$X(1.07)^5 = 60,832.64512.$$

Therefore,

$$X = \frac{60,832.64512}{(1.07)^5} \approx 43,372.83523.$$

So the amount that must be invested today is

$$\boxed{\$43,372.84}.$$